McGill initiative in Computational Medicine

# Writing with LaTeX

**Workshop Lead :** Tabish Syed

**Duration :** 2 hours (approximate)

**Registration Link :** NA

### Prerequisites

1. Web browser and internet access
2. Overleaf account.

## Summary

A hands-on introduction to LaTeX for scientific writing. Learn the basics of document preparation, how to automate cross-referencing and bibliographic citations, and how to arrange figures and visualizations for publication-ready papers.

## Learning Objectives

1. **Explain what LaTeX is**, and produce a typeset PDF document from source code using a LaTeX editor and compiler (Overleaf).
2. Build and maintain a **bibliography**: create a `.bib` database, cite sources with `\cite` commands, and format the reference list.
3. **Compose publication–quality figures**: insert, size, position, and caption images; arrange multi-panel figure grids; and cross-reference them so every figure number updates automatically.

## Content

**Part 1 — Basic Text Formatting (30 minutes)**

- `documentclass`, `sections`, `paragraphs` and `columns`
- Equations and cross-referencing

**Part 2 — Images and Figures (1 hour)**

- Single-image figure: `\includegraphics`, the `figure` environment, figure placement
- Multi-panel figures:
  - Grid of images with `tabular`
  - Grid of images using `subfigure`
  - Grid of images with `picture`/`put`

**Part 3 — Drawings with TikZ (30 minutes)**

- Basic `\draw` command: lines, arrows, simple shapes, line width and color
- Annotating images with TikZ

## Part 4 — Annotating Equations and Chemical Structures (Optional)